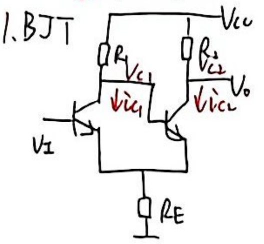
脉冲电路

一、施密特触发电路

1.BJT

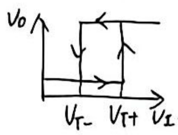


$V_I\uparrow \to i_{C1}\uparrow \to V_{C1}\downarrow \to i_{C2}\downarrow$

$\uparrow\!\!\!\!\leftarrow V_{BE}\uparrow \leftarrow V_E\downarrow$

$V_I\downarrow \to i_{C1}\downarrow \to V_{C1}\uparrow \to i_{C2}\uparrow$

$\uparrow\!\!\!\!\leftarrow V_{BE}\downarrow \leftarrow V_E\uparrow$

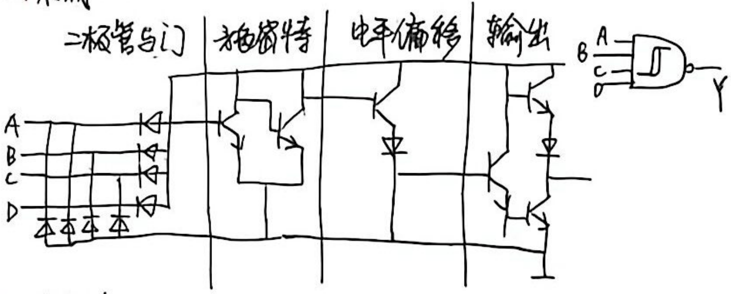


$R_1 > R_2$

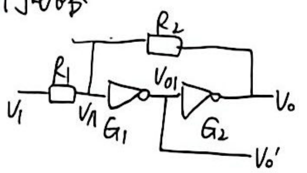
T_1饱和导通时 V_E 低于 T_2

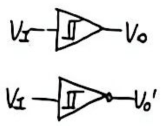
故 T_1 由截止变导通的 V_I 高于导→截

2、集成…



3、门电路



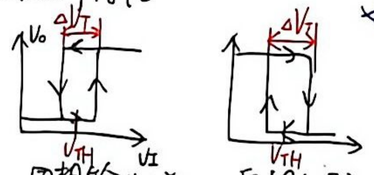


$V_A\uparrow \to V_{O1}\downarrow \to V_O\uparrow$

$$\boxed{\begin{aligned}V_{T-} &= \left(1-\frac{R_1}{R_2}\right)V_{TH}\\ V_{T+} &= \left(1+\frac{R_1}{R_2}\right)V_{TH}\end{aligned}}\qquad \Delta V_T = 2\frac{R_1}{R_2}V_{TH}$$

$R_1 < R_2$

电压传输特性



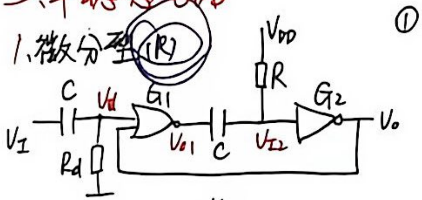
同相输出(逆)　反相(顺)

4、应用

波形变换+脉冲整形+脉冲鉴幅

二、单稳态电路

1、微分型 (R)



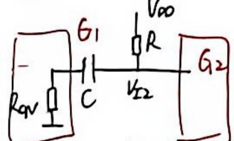
① $V_I\to 1$

$V_d\uparrow \to V_{O1}\downarrow \to V_{I2}\downarrow \to V_O\uparrow$

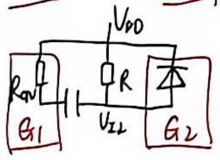
V_{O1}低 C充电至 $V_{I2}=V_{TH}$

$V_{I2}\uparrow \to V_O\downarrow \to V_{O1}\uparrow$

充



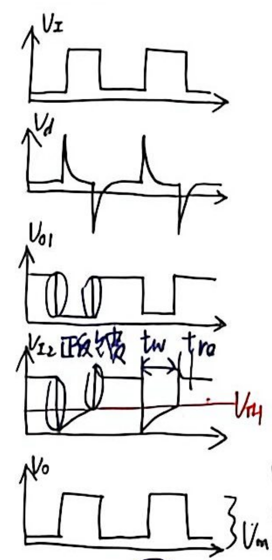
放



② $V_I\to 0$ V_{O1} V_{I2}高 $V_O=0$ C放电

稳态 $V_I=0$ $V_{I2}=V_{DD}$

$V_O=0$ $V_{O1}=V_{DD}$



$V_m = V_{OH}-V_{OL}$

微分型由于有正反馈

C充电到 V_{TH} 就会上跳到 V_{DD}

而积分型没有

C会一直放电直到 $V_I\to 0$

恢复也更慢

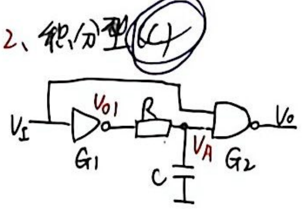
$\boxed{t_d = t_W + t_{re}}$

$$t = RC\ln\frac{V_C(\infty)-V_C(0)}{V_C(\infty)-V_{TH}}$$

$$\boxed{t_W = RC\ln\frac{V_{DD}-0}{V_{DD}-V_{TH}} = 0.69RC}$$

$\boxed{t_{re}\approx(3\sim5)R_{ON}C}$

2、积分型 (C)

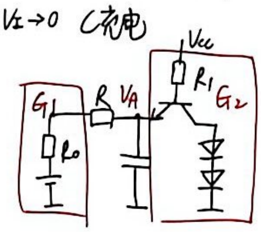


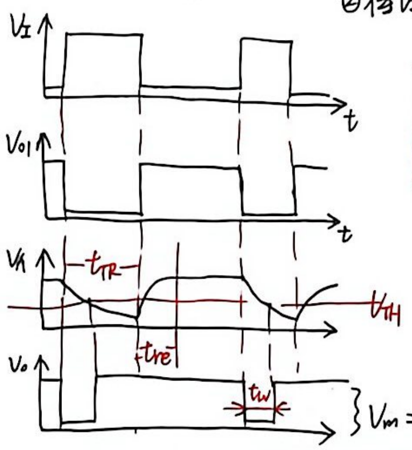
稳态 $V_I=0$ $V_O=V_{OH}$ $V_A=V_{O1}=V_{OH}$

① $V_I\to 1$ $V_{O1}\to 0$ $V_O\to 0$

C放电至 $V_A=V_{TH}$ $V_O\to 1$

② 待 $V_I\to 0$ C充电



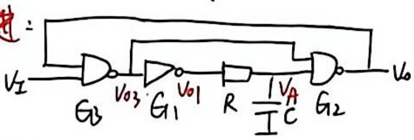


$$t_W = (R+R_O)C\ln\frac{V_{OL}-V_{OH}}{V_{OL}-V_{TH}}$$

$V_m = V_{OH}-V_{OL}$　$t_{TR}>t_W$才行

电路分辨时间 $\boxed{t_d = t_{TR}+t_{re}}$　$\boxed{t_{re}=(3\sim5)(R+R_O')C}$

改进：



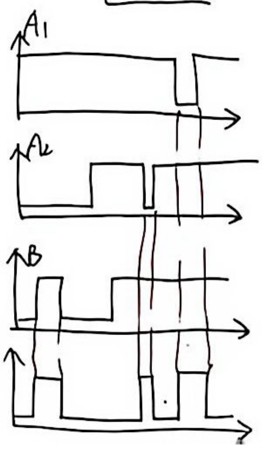
窄脉冲可触发

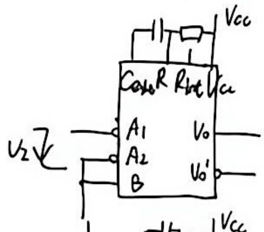
3. 集成…(74121)

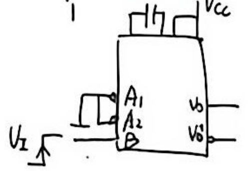
上升沿触发：B脉冲入 A_1A_2至少一个0

下降沿…：A_1/A_2脉冲入 B=1

(另一个1)



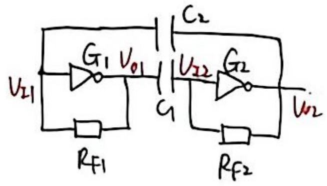




$t_W = R_{ext}C_{ext}\ln 2 = 0.69R_{ext}C_{ext}$

三、多谐振荡电路

1、对称式



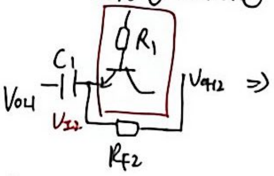
$R_{E1} = \frac{R_1 R_{F2}}{R_1 + R_{F2}}$

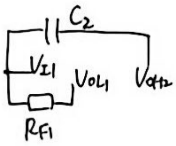
$V_{E1} = V_{OH} + \frac{R_{F1}}{R_1 + R_{F2}}(V_{CC} - V_{OH} - V_{BE})$

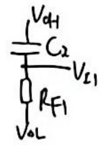
① $V_{I1}\uparrow \rightarrow V_{O1}\downarrow \rightarrow V_{I2}\downarrow \rightarrow V_{O2}\uparrow$

$V_{O2}\rightarrow 0 \quad V_{I2}\rightarrow 1$ 第一个暂稳态

C_1充电 C_2放电







C_1有两条支路充电更快

V_{I2}先上升到V_{TH} → ② $V_{I2}\uparrow \rightarrow V_{O2}\downarrow \rightarrow V_{I1}\downarrow \rightarrow V_{O1}\uparrow$

$V_{O2}\rightarrow 0 \quad V_{O1}\rightarrow 1$ 第二个暂稳态

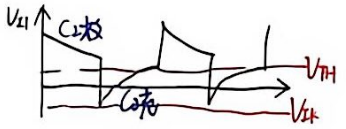
C_1放电 C_2充电

③ V_{I1}先上升到V_{TH} ⇒ ①

$V_{OL} \approx 0 \quad V_{C1}(0) = V_{IK}$

$V_{C1}(\infty) = V_{E1}$

$T_1 = R_{E1} C_1 \ln \frac{V_{E1} - V_{IK}}{V_{E1} - V_{TH}}$



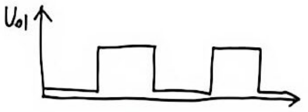
$$\boxed{T = 2T_1 = 2R_E C \ln \frac{V_E - V_{IK}}{V_E - V_{TH}}}$$

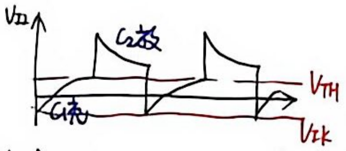
V_{IK}为TTL输入端反向钳位

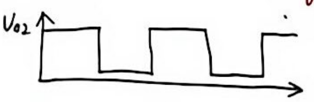
取$V_{OH} = 3.4V \quad V_{IK} = -1V \quad V_{TH} = 1.1V$

在$R_F \ll R_1$时

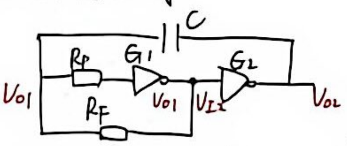
$$\boxed{T \approx 1.3 R_F C}$$







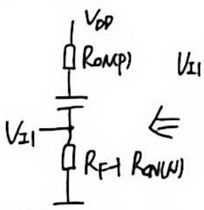
2、非对称式

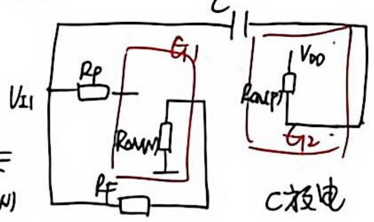


① $V_{I1}\uparrow \rightarrow V_{I2}\downarrow \rightarrow V_{O2}\uparrow$

$V_{O1}\rightarrow 0 \quad V_{O2}\rightarrow 1$ 第一个暂稳态

$R_F \gg R_{ON}$



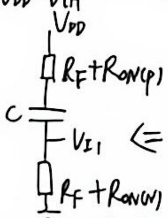


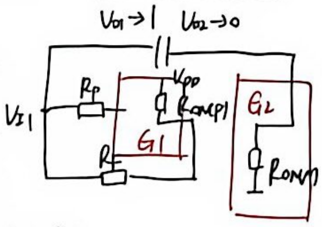
C放电

$T_1 = R_F C \ln \frac{V_{DD} - (V_{TH} - V_{DD})}{V_{DD} - V_{TH}} = R_F C \ln 3$

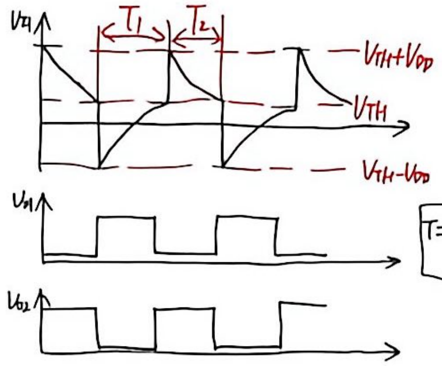
② $V_{I1}\downarrow \rightarrow V_{I2}\uparrow \rightarrow V_{O2}\downarrow$

$V_{O1}\rightarrow 1 \quad V_{O2}\rightarrow 0$





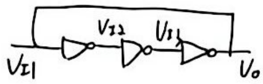
$T_2 \approx R_F C \ln \frac{0 - (V_{TH} + V_{DD})}{0 - V_{TH}} = R_F C \ln 3$



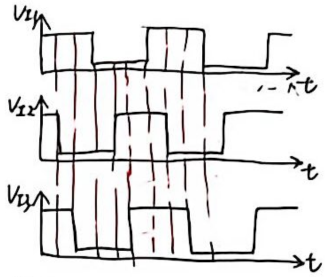
$$\boxed{T = T_1 + T_2 \approx 2R_F C \ln 3 = 2.2R_F C}$$

3、环形振荡器

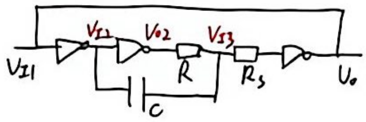
① 最简单



$T = 2n\,t_{pd}$

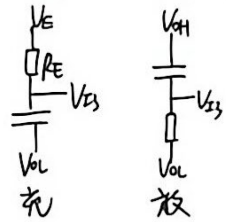


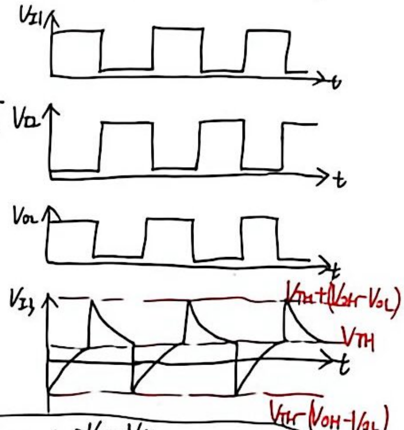
② 带RC延迟



若$R + R_S \gg R \quad V_{OL} \approx 0$

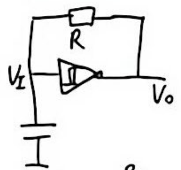
则$V_E \approx V_{OH} \quad R_E \approx R$

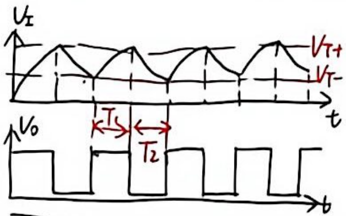


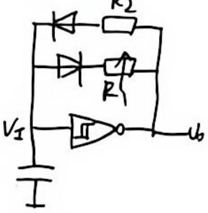


$$\boxed{T_1 \approx RC \ln \frac{2V_{OH} - V_{TH}}{V_{OH} - V_{TH}} \quad T_2 \approx RC \ln \frac{V_{OH} + V_{TH}}{V_{TH}} \quad T = T_1 + T_2 \approx 2.2RC}$$

4、施密特构成

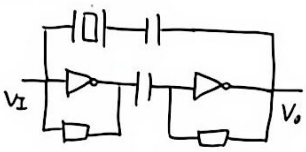






$$\boxed{T = T_1 + T_2 = RC \ln \frac{V_{DD} - V_{T-}}{V_{DD} - V_{T+}} + RC \ln \frac{V_{T+}}{V_{T-}}}$$

5、石英晶体



① 工作过程，波形，关键电压

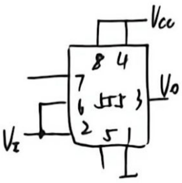
② 充放电等效电路

③ 修订关键值，算充放电时间

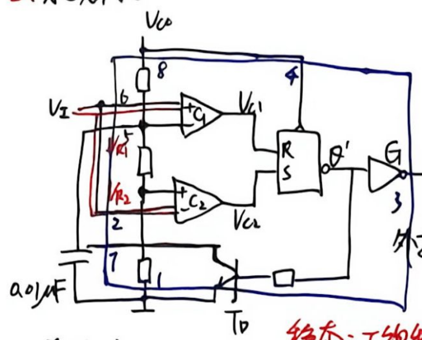
四、555定时器

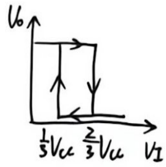
1、功能表

R_D'	V_{I1}(TH) (6)	V_{I2}($\overline{TR'}$) (2)	V_0 (3)	T_D (7)
0	X	X	低	通
1	$>\frac{2}{3}V_{cc}$	$>\frac{1}{3}V_{cc}$	低	通
1	$<\frac{2}{3}V_{cc}$	$>\frac{1}{3}V_{cc}$	不变	不变
1	$<\frac{2}{3}V_{cc}$	$<\frac{1}{3}V_{cc}$	高	止
1	$>\frac{2}{3}V_{cc}$	$<\frac{1}{3}V_{cc}$	高	止



2、施密特





$\Delta V_T=\frac{1}{3}V_{cc}$

外接参考

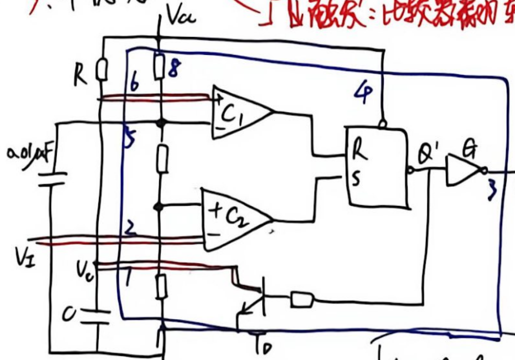
$V_{T+}=V_{CO}$　$V_{T-}=\frac{1}{2}V_{CO}$

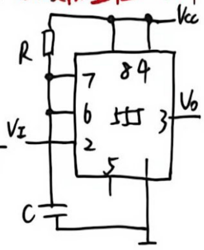
$\Delta V_T=\frac{1}{2}V_{CO}$

3、单稳态

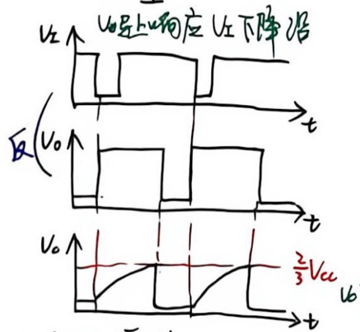
稳态：T饱和导通 $V_o=0$ → T截止

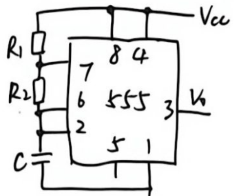
触发：比较器翻转→触发器置位→$V_o=1$





$$t_w=RC\ln\frac{V_{cc}-0}{V_{cc}-\frac{2}{3}V_{cc}}=RC\ln 3=1.1RC$$

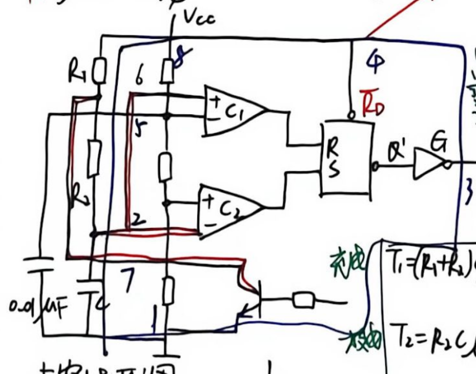


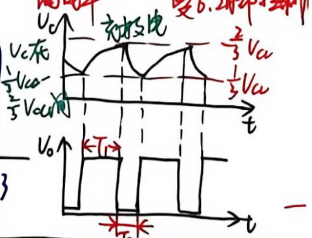


4、多谐振荡

当4为低电平时，输出被置0

高电平 受6,2脚控制





充电　$T_1=(R_1+R_2)C\ln 2=(R_1+R_2)C\ln\frac{V_{cc}-V_{T-}}{V_{cc}-V_{T+}}$

放电　$T_2=R_2C\ln 2=R_2C\ln\frac{0-V_{T+}}{0-V_{T-}}$

$q=\frac{T_1}{T}=\frac{R_1+R_2}{R_1+2R_2}$　占空比

设 $C=100\mu F$

占空比可调